POWER BI | ITSM ANALYTICS | CASE STUDY

IT Service Operations Control Dashboard

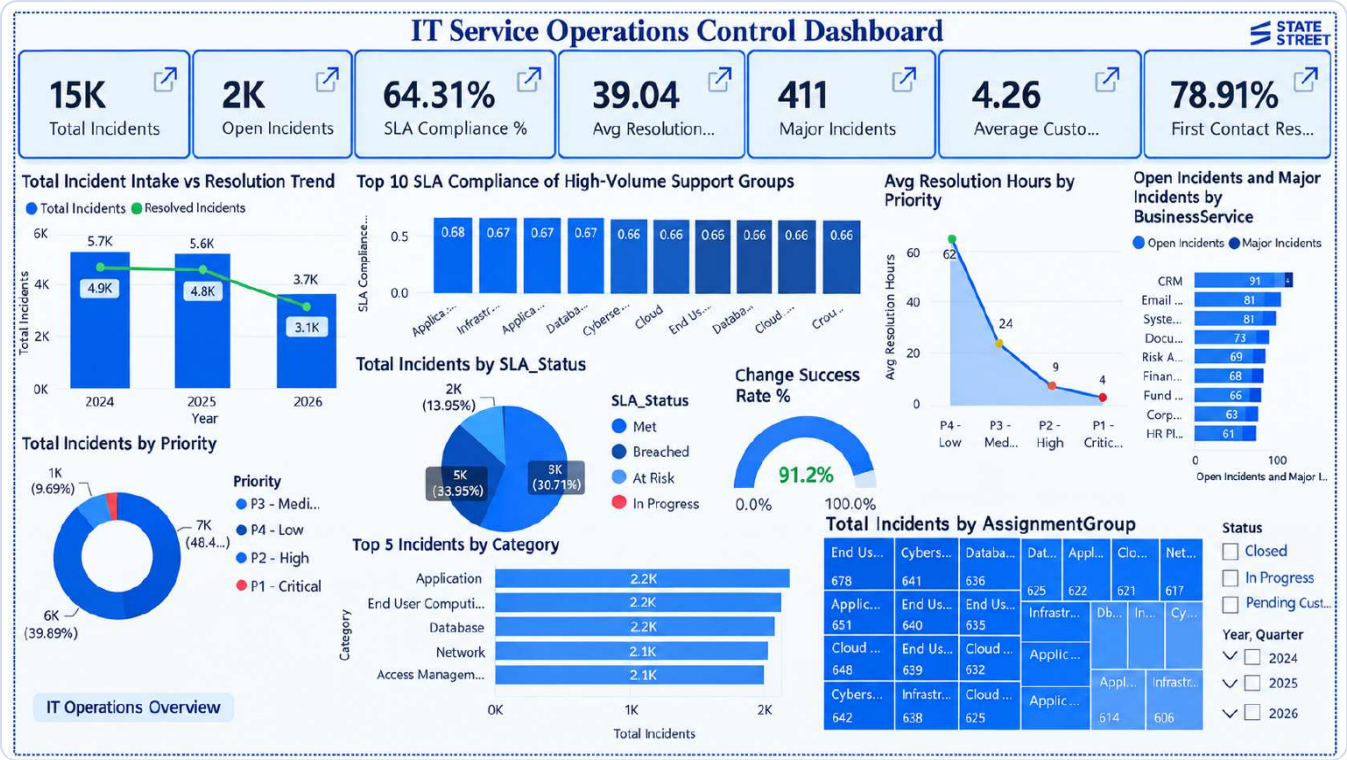
An 8-page Power BI solution that turns 15,000 incident records and 2,000 change records into one decision-ready view of SLA, risk, resolution speed and customer experience.

Power BI | DAX | Power Query | Data Modeling | ITSM / ITIL

15K	2.1K	64.31%	39.04h	411	4.26	78.91%
Total Incidents	Open Incidents	SLA Compliance	Avg Resolution	Major Incidents	Avg CSAT	First Contact Res.

Project snapshot	
Domain	Financial-services IT operations (ITSM): incident, change and service-level management
Objective	Give IT leaders one self-service view of volume, SLA health, open-incident risk, major incidents, change quality and customer satisfaction
Scope	15,000 incidents and 2,000 changes, 2024-2026, 5 regions, 6 towers, 8 requester departments, 4 priority levels
Deliverable	8-page interactive Power BI report with drill-through navigation, slicers and 40+ visuals
Tech stack	Power BI Desktop, Power Query (M), DAX, star-schema data model
Project type	Portfolio / analytics case study

DASHBOARD PREVIEW | Page 1: IT Operations Overview



01 BUSINESS CONTEXT

Problem, objectives and solution architecture

IT service desks generate thousands of tickets a month, but SLA performance, open-ticket risk, major outages, change failures and customer sentiment usually sit in separate reports. Leaders cannot see **where service is breaking down** or **which teams need help** without manual consolidation. This project unifies those views into a single, filterable control room.

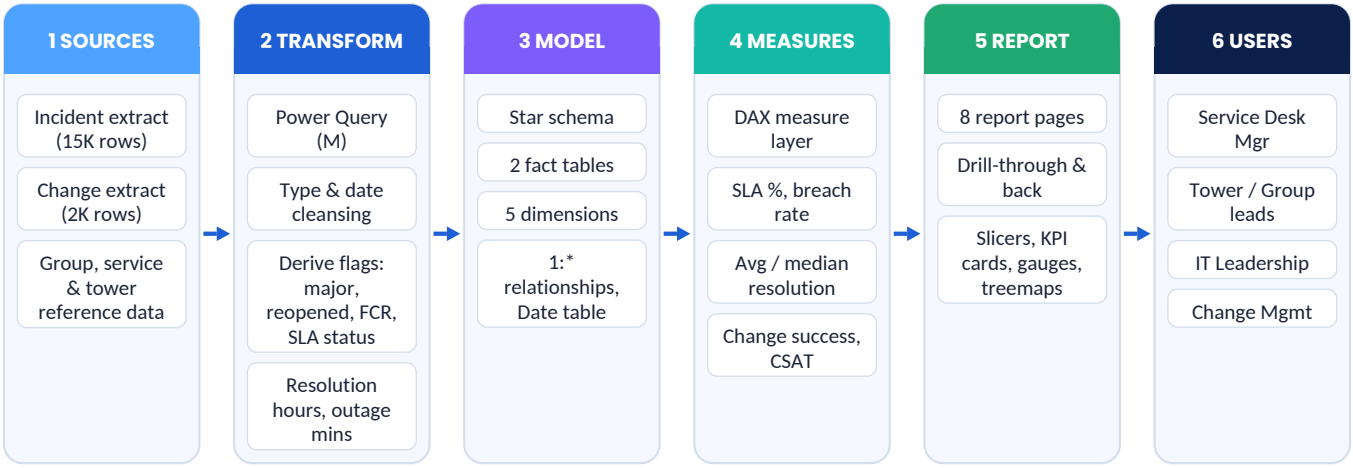
The challenge

- SLA, incident and change data tracked in separate reports
- Slow, manual root-cause analysis for breaches and reassignments
- No single view of open risk (P1/P2) by team and business service
- Customer experience not linked to operational performance

Objectives

- Track **SLA compliance** and breach rate by priority, group and month
- Expose **open and at-risk incidents** for fast triage
- Analyse **major incidents** by region, tower, category, manager
- Measure **resolution speed**, reassignment and outage impact
- Link **CSAT and first-contact resolution** to support groups
- Monitor **change success**, rollbacks and emergency changes

Solution architecture



Delivery lifecycle



ITSM process coverage

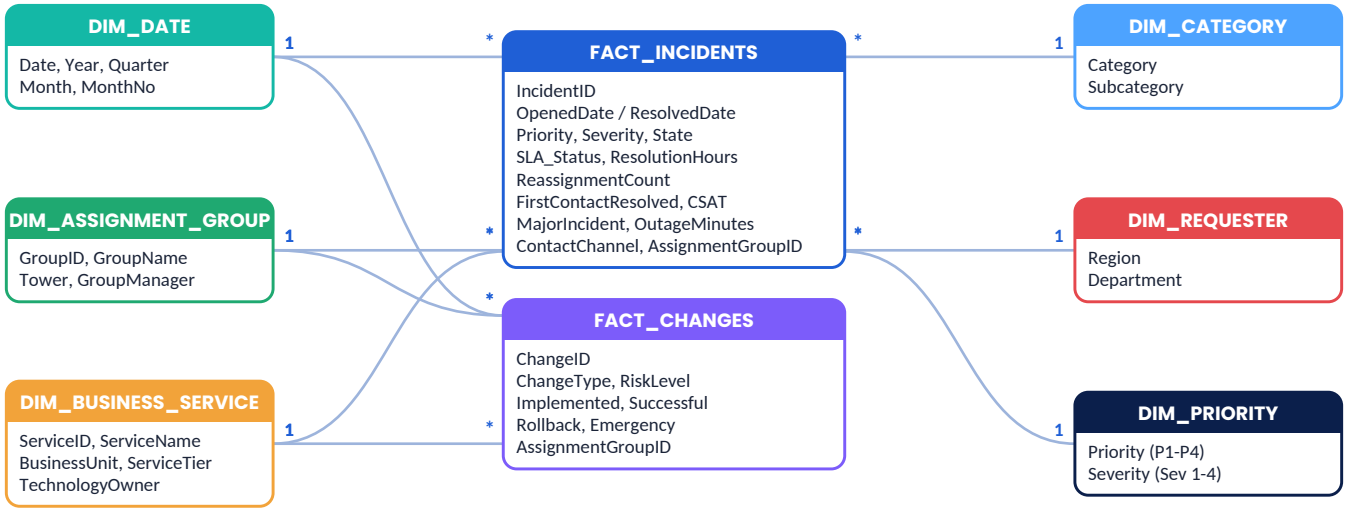
ITIL practice	Questions answered	Dashboard page
Incident management	How many, how severe, where from, via which channel?	Overview, Total Incident
Service level management	Are we meeting SLA, and for which priorities?	SLA Analysis
Major incident management	Which regions, towers and managers carry outages?	Major Incident
Change enablement	Are changes succeeding, rolling back or going emergency?	Change Management
Service desk and CX	Is first-contact resolution turning into satisfied users?	Customer Satisfaction

Design principle
One consistent layout across all pages: KPI strip on top, trend and breakdown visuals in the middle, slicers on the right, back and drill-through buttons top-left.

02 DATA MODEL AND MEASURES

Star schema, data dictionary and DAX logic

The model separates **what happened** (facts) from **how we slice it** (dimensions). Shared dimensions let one slicer filter both incidents and changes, which keeps every page consistent.



Logical model reflecting the fields used across the report. Relationships are one-to-many, single direction from dimension to fact.

Key DAX measures

measures.dax

```
// SLA
SLA Compliance % = DIVIDE( [SLA Met Incidents], [SLA Met Incidents] + [SLA Breached Incidents] )
SLA Breach Rate % = 1 - [SLA Compliance %]
// Speed & efficiency
Avg Resolution Hours = AVERAGE( Incidents[ResolutionHours] )
Median Resolution Hours = MEDIAN( Incidents[ResolutionHours] )
Reassignment Rate % = DIVIDE( CALCULATE( COUNTROWS(Incidents), Incidents[ReassignmentCount] > 0 ), [Total Incidents] )
// Risk & experience
Open Incidents = CALCULATE( [Total Incidents], Incidents[State] <> "Closed", Incidents[State] <> "Resolved" )
FCR % = DIVIDE( CALCULATE( [Total Incidents], Incidents[FirstContactResolved] = TRUE() ), [Total Incidents] )
Change Success Rate % = DIVIDE( [Successful Changes], [Implemented Changes] )
```

Data preparation and quality checks

- **Power Query:** typed all date and numeric columns, standardised priority and severity labels, removed duplicates and blank keys.
- **Derived columns:** major-incident flag, reopened flag, resolution hours, outage minutes, SLA status.
- **Reconciliation:** totals tie across pages (Total 15,000 = Resolved 12,905 + Open 2,095; SLA Met 8,299 + Breached 4,606 = 12,905).
- **Governance:** consistent measure names, a display-folder per business area, hidden technical keys.

Data dictionary: key fields

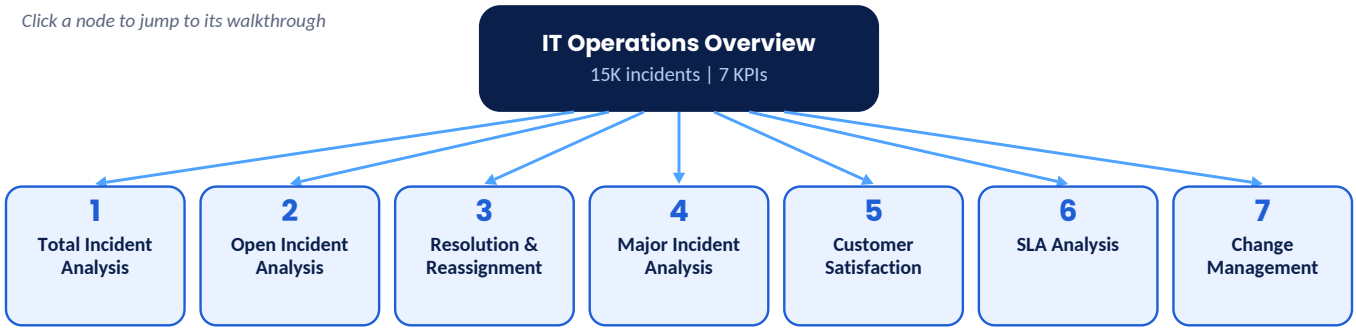
Field	Definition	Used in
SLA_Status	Met, Breached, At Risk or In Progress, based on target vs actual resolution	SLA, Overview
ResolutionHours	Hours from opened to resolved timestamp	Resolution, Overview
MajorIncident	Flag for P1/Sev 1 style outages with business-wide impact	Major Incident
FirstContactResolved	True when resolved at first contact without reassignment	Customer Satisfaction
RollbackFlag / EmergencyFlag	Change reverted after implementation / raised outside normal window	Change Management

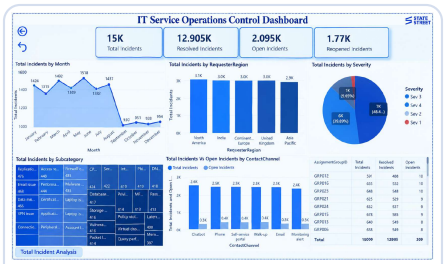
03 DASHBOARD WALKTHROUGH

Navigation design and page-by-page guide

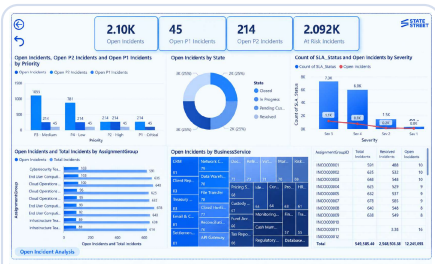
The report is built as a **hub-and-spoke**: an executive overview links to seven analytical pages. Every spoke page carries Back and Reset buttons so users never get lost, and slicers stay in the same position everywhere.

Click a node to jump to its walkthrough





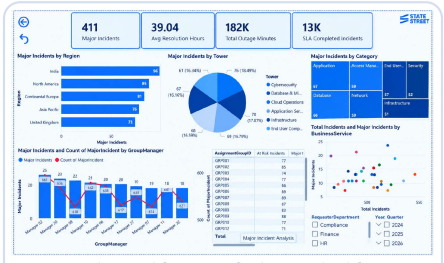
1 Total Incident Analysis: monthly trend, region, severity, subcategory, channel



2 Open Incident Analysis: open P1/P2, at-risk, by group and service



3 Resolution & Reassignment: avg/median hours, outage, reassignment rate



4 Major Incident Analysis: 411 incidents by region, tower, category, manager



5 Customer Satisfaction: CSAT 4.26, FCR 78.9%, satisfaction rate 71.9%



6 SLA Analysis: 64.31% compliance, by priority and over time



7 Change Management: 2K changes, rollbacks, emergency changes, tower view

Reading the pages

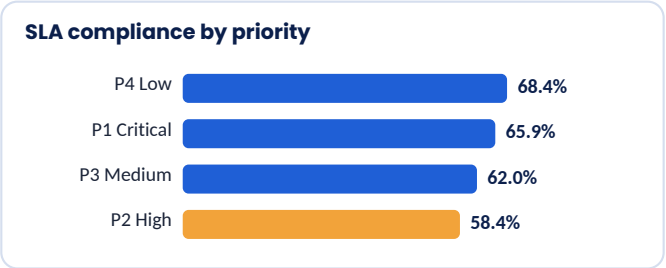
Overview is page 1. Each of the 7 pages answers one question: Where is volume? What is open? How fast? What is major? Are users happy? Are we on SLA? Are changes safe?

Navigation and UX features

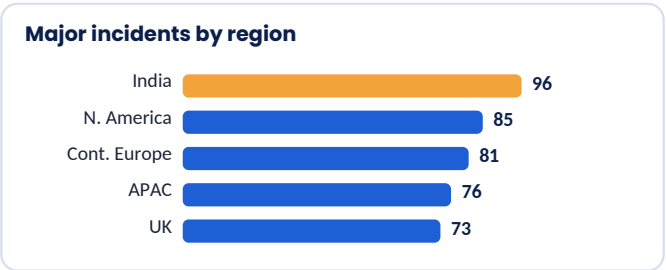
Feature	How it helps
Back and Reset buttons (top-left)	One-click return to the overview and clearing of all filters
Consistent slicers	Year/Quarter, Priority, Severity, Service Tier and Risk Level sit in the same place on every page
KPI cards on every page	Each page opens with 4 headline numbers before any detail
Cross-filtering	Selecting a bar, tower or group filters every other visual on the page

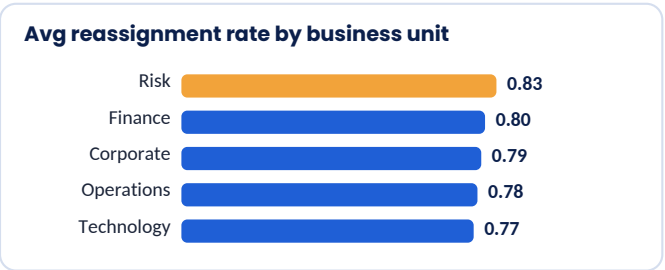
04 INSIGHTS AND OUTCOMES

What the data says and what to do about it









Key findings

- **SLA gap:** 64.31% compliance overall; **P2-High is weakest at 58.43%**, below P1-Critical (65.88%). High-priority work is not getting the attention its label implies.
- **Open-risk concentration:** 2.1K open incidents, including **45 P1 and 214 P2**; almost all open items are flagged at risk of breach.
- **Reassignment drag:** a 78.29% average reassignment rate, highest in Risk (0.83), is a likely driver of the 39-hour average resolution time (median 26.84h shows a long-tail of slow tickets).
- **Major incidents:** 411 incidents and about 182K outage minutes (about 3K hours); India (96) leads by region and Cybersecurity (76) by tower.
- **Experience:** CSAT holds at 4.26 and FCR at 78.9%, with the monthly satisfaction rate ranging from 69.8% (August) to about 72%.
- **Change quality:** 122 rollbacks and 158 emergency changes out of 2,000; change success rate stands at 91.2%.

Recommendations

Focus area	Recommended action	KPI to watch
P2-High SLA	Add queue-aging alerts and an escalation rule at 70% of SLA time	P2 SLA compliance %
Reassignments	Improve routing rules and knowledge articles for groups above 0.80	Reassignment rate %
Open P1/P2	Daily triage on at-risk list by group and business service	Open P1 / P2 count
Major incidents	Run problem reviews for India and Cybersecurity hotspots	Major incidents, outage hrs
Change risk	Review emergency changes and rollbacks by tower	Change success %, rollbacks

Skills demonstrated

Power BI | DAX | Power Query | Star-schema modeling | KPI design | ITSM / ITIL metrics | Drill-through UX | Data validation | Storytelling

Business value

A single control room replaces five manual reports, shortens triage, and lets leaders move from *what happened* to *who should act* in two clicks.